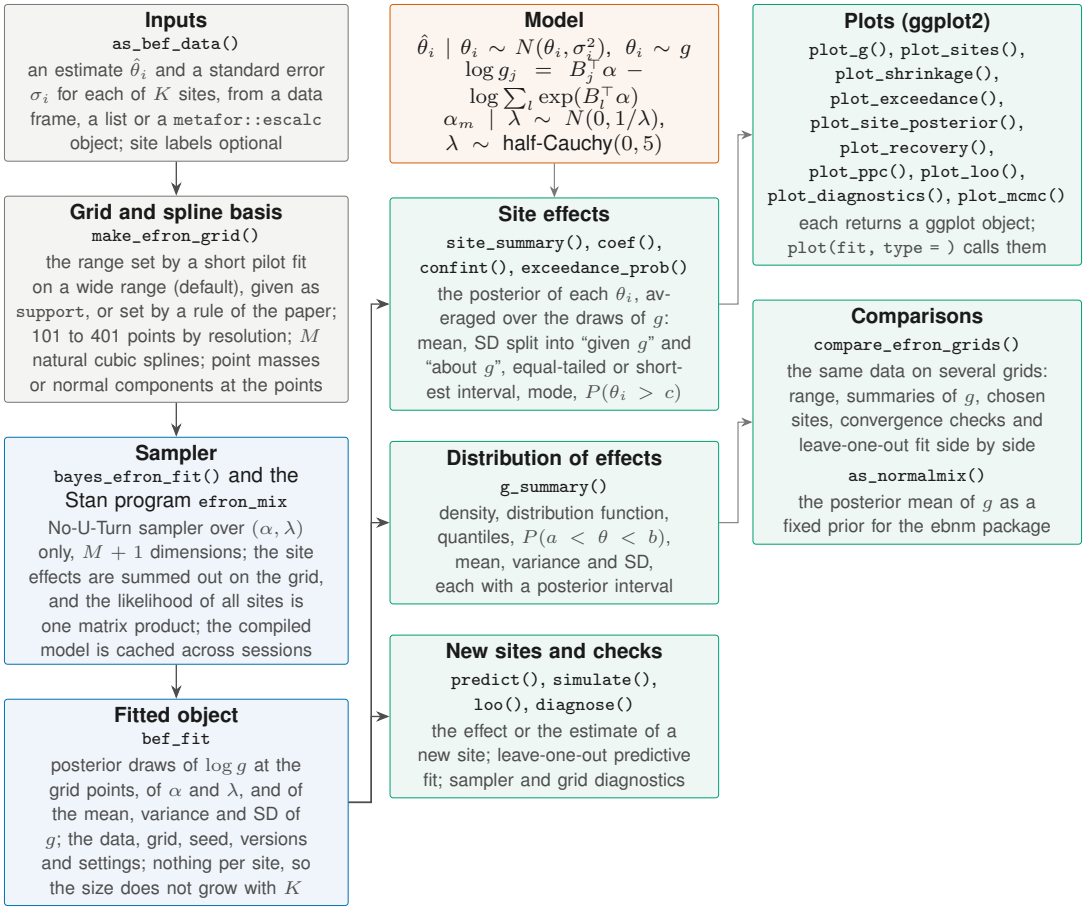

Inputs
as_bef_data()
an estimate $\hat{\theta}_i$ and a standard error σ_i for each of K sites, from a data frame, a list or a metafor::escalc object; site labels optional
Grid and spline basis
make_efron_grid()
the range set by a short pilot fit on a wide range (default), given as support, or set by the rule of the paper; 101 to 401 points by resolution; M natural cubic splines; point masses or normal components at the points
Sampler
bayes_efron_fit() and the Stan program efron_mix
No-U-Turn sampler over (α, λ) only, $M + 1$ dimensions; the site effects are summed out on the grid, and the likelihood of all sites is one matrix product; the compiled model is cached across sessions
Fitted object
bef_fit
posterior draws of $\log g$ at the grid points, of α and λ, and of the mean, variance and SD of g; the data, grid, seed, versions and settings; nothing per site, so the size does not grow with K
Model
$\hat{\theta}_i \mid \theta_i \sim N(\theta_i, \sigma_i^2),\ \theta_i \sim g$
$\log g_j = B_j^\top \alpha - \log \sum_l \exp(B_l^\top \alpha)$
$\alpha_m \mid \lambda \sim N(0, 1/\lambda),$
$\lambda \sim$ half-Cauchy$(0, 5)$
Site effects
site_summary(), coef(), confint(), exceedance_prob()
the posterior of each θ_i, averaged over the draws of g: mean, SD split into "given g" and "about g", equal-tailed or shortest interval, mode, $P(\theta_i > c)$
Distribution of effects
g_summary()
density, distribution function, quantiles, $P(a < \theta < b)$, mean, variance and SD, each with a posterior interval
New sites and checks
predict(), simulate(), loo(), diagnose()
the effect or the estimate of a new site; leave-one-out predictive fit; sampler and grid diagnostics
Plots (ggplot2)
plot_g(), plot_sites(), plot_shrinkage(), plot_exceedance(), plot_site_posterior(), plot_recovery(), plot_ppc(), plot_loo(), plot_diagnostics(), plot_mcmc()
each returns a ggplot object; plot(fit, type =) calls them
Comparisons
compare_efron_grids()
the same data on several grids: range, summaries of g, chosen sites, convergence checks and leave-one-out fit side by side
as_normalmix()
the posterior mean of g as a fixed prior for the ebnm package